

WORKFLOW & INTEGRATION SPECIFICATION

SENTINEL AI SURVEILLANCE PLATFORM

End-to-End Operational Pipeline & Police Integration Interfaces

Detailed sequence diagrams, glass-to-alert latency budgets, WebSocket event schemas, PostGIS spatial queries, and CCTNS hotlist sync specifications.

Document Code:

SENTINEL-INT-v1.0

Target Agency:

Gujarat Police Command Center

Pipeline Latency:

< 400ms Glass-to-Alert

Message Broker:

Redis Pub/Sub + WebSockets

1. 5-Stage Sequential Operational Pipeline

Stage 1: Camera Ingestion & Motion Sampling (0 – 40ms)

Camera streams H.264/H.265 video over RTSP. MediaMTX proxies the stream while the adaptive frame grabber samples frames (10 FPS) when motion is detected.

Stage 2: AI ANPR Detection & OCR (40 – 195ms)

YOLOv8 isolates the vehicle and license plate bounding boxes (45ms). CLAHE enhancement sharpens the crop, and PaddleOCR PP-OCRv4 decodes the characters (110ms).

Stage 3: Indian Syntax Validation & Redis Match (195 – 202ms)

Regex disambiguation repairs OCR optical ambiguities. Redis executes an in-memory hash check against active stolen/wanted lists in under 1ms.

Stage 4: PostGIS Spatial Storage & Speed Calc (202 – 230ms)

Coordinates and detection data are stored in PostGIS. The trajectory engine calculates delta-T transit time and average speed from the previous camera node.

Stage 5: Real-Time WebSocket Siren Broadcast (230 – 260ms)

Async WebSocket emits a JSON alert payload to operator dashboards, triggering the audio siren, red banner, and GIS radar pulse.

2. Glass-to-Alert Latency Budget

Pipeline Stage	Operation	GPU Latency (RTX 4060)	CPU Latency (Intel i7)
1. Ingestion	RTSP Grab & Decode	35 ms	50 ms
2. YOLOv8	Vehicle & Plate Localization	45 ms	90 ms
3. PaddleOCR	PP-OCRv4 Text Recognition	65 ms	120 ms
4. Redis Match	In-Memory Hotlist Check	1 ms	2 ms
5. PostGIS & WS	Spatial Write & Alert Push	25 ms	40 ms
Total End-to-End	Optical Capture to Screen Siren	~171 ms	~302 ms

3. WebSocket Live Alert Payload Schema

```
{
  "event_type": "HOTLIST_ALERT",
  "timestamp": "2026-08-31T10:45:22.108Z",
  "camera": {
    "id": "CAM-AHM-04",
    "name": "SG Highway - Pakwan Cross Road",
    "coordinates": { "latitude": 23.0338, "longitude": 72.5074 }
  },
  "detection": {
    "license_plate": "GJ01AB1234",
    "vehicle_class": "SUV (Fortuner)",
    "ocr_confidence": 0.964,
    "speed_estimate_kmh": 58.4,
    "crop_url": "/api/v1/media/crops/GJ01AB1234.jpg"
  },
  "hotlist_match": {
    "severity": "CRITICAL",
    "category": "STOLEN_VEHICLE",
    "fir_number": "FIR-2026-9081",
    "police_station": "Vastrapur PS, Ahmedabad"
  }
}
```

4. External Police Integrations

CCTNS Integration: Automated nightly CSV/JSON sync or real-time webhook ingestion for active FIR stolen vehicle lists.

PCR Intercept Dispatch: Webhook dispatches GPS coordinates and fleeing heading to the nearest mobile patrol unit.